

A Cooperative Signalling-Game Formulation for Multi-Agent Highway Driving

Measuring Emergent Behavioural Conventions via
Multi-Agent Reinforcement Learning

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Abstract

Autonomous highway driving is, at its core, a multi-agent decision problem: every vehicle carries a private behavioural type that can't be directly observed. Two broad approaches have dominated the field. The first infers types from observed trajectories but doesn't account for the fact that the ego-vehicle is simultaneously being inferred by others. The second solves analytical equilibria under assumed payoffs with hand-specified signals. Neither asks what the classical signalling-game literature treats as primary - given a shared behavioural signal space, what communicative conventions actually emerge from interaction?

This thesis takes that question seriously. Cooperative multi-agent highway driving is formulated as a repeated signalling game over a small discrete vocabulary of behavioural signals whose meanings aren't pre-assigned, and the methodological machinery is developed to measure emergent informativeness via multi-agent reinforcement learning with a recurrent type encoder. Three concrete deliverables follow: (i) a formal cooperative signalling-game model for continuous-control highway driving with a behaviourally-grounded, non-cheap-talk signal space; (ii) an operational mutual-information criterion, ρ , for detecting pooling, partial-pooling, and separating regimes; and (iii) an experimental demonstration across three conditions establishing that ρ responds to the behavioural content of type labels rather than their mere presence, and that the predominant pooling at the current preference divergence ($d = 0.30$) is consistent with the Spence/Crawford-Sobel prediction that separation requires $d > d_{\text{sep}}$, a threshold derived analytically from the reward structure's single-crossing condition.

1 Introduction

Highway driving means constant negotiation with vehicles you can't fully read. Caution, assertiveness, risk tolerance, preferred following distance - none of it is directly observable. The field's standard answer has been perception, prediction, planning: build a probabilistic model of neighbour motion and let the planner marginalise over it [2, 17]. Prediction methods have come a long way - from rule-based forward simulation, through belief tracking over discrete intent hypotheses [1, 19], to unsupervised latent-trait inference with recurrent or attention-based encoders [8, 9, 11].

But that entire trajectory frames the problem one-sidedly. The ego-vehicle asks *what will my neighbour do?* and stops there. A human driver also asks the other half: *what do my neighbours think I am, and is my behaviour shaping that belief?* Real drivers don't just execute manoeuvres at minimum cost. The way they close a gap, ease off the accelerator, or begin a lane change typically carries more information than what's strictly needed to complete the move. Those excess margins are signals, and other drivers read them [5, 12, 15].

Consider a concrete two-vehicle instance. Car A is cautious; car B is assertive. Both types are private: neither vehicle can directly observe the other's behavioural disposition.

B approaches A from behind in the same lane. Over the preceding five seconds, B has maintained a tight following distance, accepted small gaps, and initiated accelerations sharply. A observes this kinematic history and updates its belief: the follower is probably assertive. A responds by holding a wider headway than it otherwise would, smoothing its own deceleration profile to avoid provoking a rear-end conflict. Critically, A did not change its *destination* - it is still tracking the same speed target - but it changed *how* it pursues that target, because its model of B has changed.

Meanwhile, B is reading A. A's wide margins, early braking, and smooth acceleration smoothness are recognised as the kinematic signature of a cautious type. B updates its belief accordingly and adjusts its gap-acceptance threshold upward: it will wait for a larger gap before forcing a merge, because it has learned that A will not accelerate assertively to close one. Again, B's destination is unchanged; only its execution style shifts in response to its inferred model of A.

Neither vehicle has sent an explicit message. Coordination emerged entirely from the observable style of otherwise routine manoeuvres. The same physical action - maintain speed, decelerate, drift laterally - carried different information depending on *how* it was executed, and both vehicles exploited that information to reduce conflict without any direct communication channel.

This is the province of signalling-game theory [3, 14]. Informed senders transmit observable signals; uninformed receivers update beliefs via Bayes' rule and respond accordingly. Equilibria fall into three canonical regimes: *separating*, where different types emit different

signals and communication is fully informative; *pooling*, where all types converge on one signal and communication collapses; and *partial-pooling* somewhere in between [6]. The existence of a signal space doesn't determine by itself whether it'll be used informatively. Whether it is depends on the game structure.

This thesis casts cooperative multi-agent highway driving as such a signalling game, then builds the empirical tools to determine which regime actually emerges when agents are trained over a shared discrete vocabulary whose meanings are deliberately left unassigned. The cooperative-population restriction is methodologically deliberate: it removes strategic deception as a confound. Any pooling outcome can't be attributed to agents intentionally misrepresenting themselves, so the presence or absence of separation becomes diagnostic of game structure rather than adversarial strategy.

2 Research Questions

RQ1 - Measurable emergence of non-trivial signal structure. When cooperative agents are trained jointly with a shared discrete behavioural signal space over a heterogeneous population, does the separation coefficient ρ (Section 4) reliably exceed a random-signal baseline across training seeds?

RQ2 - Necessity of type heterogeneity. Does ρ differ between a bimodal population where type labels correspond to distinct reward functions and one where both labels share identical reward weights? In the same-reward control, both type tokens are present and sampled equally so ρ remains well-defined, but the labels carry no behavioural content - no incentive gradient exists to drive separation. If ρ stays near the random-signal level in this control but rises in the bimodal condition, any elevation is attributable to the behavioural content of the type labels rather than to architectural effects of the signalling head.

3 Method

3.1 Formalism

The highway is modelled as a partially observable stochastic game. Each agent i has a latent type $\theta_i \in \Theta$, fixed within an episode and resampled across episodes from a population distribution. We use $|\Theta| = 2$ (cautious, assertive), with the cluster determining the weighting of safety, progress, and comfort terms in the agent's intrinsic reward. Types are private; there are no persistent identifiers across episodes, and no communication channel exists other than the behavioural signal space defined below.

At each decision step t , agent i selects a driving action $a_i \in \mathcal{A}$ - continuous: longitudinal acceleration, steering rate - and a discrete signal $m_i \in \mathcal{M}$ from a vocabulary of cardinality $|\mathcal{M}| = 3$. The signal is a discrete selector that determines which of three pre-specified execution profiles renders a_i . Every profile is physically feasible and driving-legal; what

differs across them is the observable kinematic signature.

A joint policy is a pair (σ_i, π_i) for each agent: a signalling strategy $\sigma_i(s_i, \theta_i) \rightarrow \Delta(\mathcal{M})$ and a response strategy $\pi_i(s_i, b_i) \rightarrow \Delta(\mathcal{A})$, where b_i is the belief held by agent i over the types of currently visible neighbours. The analytical reference point is Perfect Bayesian Equilibrium [6]: strategies are mutually best-responding, and on-path beliefs are Bayes-updated as

$$b_i(\theta_j \mid m_j, h_j) \propto \sigma_j(m_j \mid s_j, \theta_j) b_i(\theta_j \mid h_j), \quad (1)$$

where h_j is the recent observed behavioural history of agent j . We don't claim to find a PBE; we train via MARL and characterise whatever equilibrium the training procedure reaches.

3.2 Architecture

For each visible neighbour j , a type embedding is produced by a single-layer GRU encoder over a windowed sequence of observed kinematic features and emitted signals:

$$\hat{\theta}_{ij}(t) = f_\phi\left(\left\{(o_j(s), m_j(s))\right\}_{s=t-W}^t\right),$$

with window $W = 5$ s. A GRU is chosen as the primary encoder for training stability at this scale. The encoder is shared across agents under parameter sharing during centralised training; at execution time every agent runs its own copy over its own observations.

The full policy decomposes into three heads sharing the encoder. There's the type encoder f_ϕ ; a signalling head σ_θ mapping $(s_i, \theta_i, \{\hat{\theta}_{ij}\}_j)$ to a categorical distribution over $\mathcal{M}$ on the sender side; and a response head π_ψ mapping $(s_i, \{\hat{\theta}_{ij}\}_j)$ to a continuous driving action on the receiver side - this head doesn't take the agent's own type as input. Neighbour embeddings are aggregated by a permutation-invariant DeepSets layer [18] to handle variable neighbour counts. Training uses MAPPO [16] under centralised training with decentralised execution [10]. MAPPO is the natural choice here: the signalling head is categorical, and MAPPO is the established default for cooperative MARL with mixed discrete-continuous action spaces. The indirect credit-assignment problem for the signalling head — gradient arrives only through downstream neighbour behaviour — is mitigated by two mechanisms: the centralised critic provides a variance-reducing value baseline for the REINFORCE gradient through the discrete signal, and a separate entropy coefficient on the signalling head is annealed linearly from a high initial value to a low final value over training, maintaining signal exploration until type-informative structure can form.

3.3 Signal Space

$\mathcal{M}$ is a small discrete vocabulary of fixed execution profiles. For any commanded driving action a , each $m \in \mathcal{M}$ specifies a distinct but physically feasible way to realise a . Each profile is a parametric override passed to the IDM/MOBIL low-level controllers in

highway-env: the commanded action a_i sets the target, and the profile determines the kinematic envelope within which it is executed. The three profiles are defined by the parameters in Table 1.

Profile	Headway multiplier	Jerk cap (m s^{-3})	Gap-accept threshold (m)	Lateral drift at initiation (m)
m_1 (conservative)	$1.5\times$	1.0	15	0.3
m_2 (neutral)	$1.0\times$	2.0	10	0.6
m_3 (assertive)	$0.7\times$	4.0	6	1.0

Table 1: Execution-profile parameters. Headway multiplier scales the IDM desired headway; jerk cap bounds longitudinal acceleration rate; gap-accept threshold is the minimum accepted merge gap; lateral drift is the anticipatory displacement at lane-change initiation.

Two design commitments follow from this. First, signal emission is costly in the sense of Spence [14]: different profiles incur different costs across progress, comfort, and safety dimensions, and no single profile dominates for every type. This cost asymmetry is what gives a separating equilibrium its leverage - cautious types find it cheaper to emit cautious profiles, assertive types find it cheaper to emit assertive ones - and it’s exactly what a cheap-talk channel lacks. Second, and just as importantly, the semantics of each signal are deliberately left unassigned. No profile is labelled as meaning *I am cautious*. Whether m_k comes to denote a particular type is an emergent property of the trained population, measured rather than stipulated.

3.4 Reward Structure

The reward for agent i is $R_i = R_i^{\text{drive}}$, the standard driving reward over forward progress, collision avoidance, lane-keeping, and comfort, weighted according to θ_i [7]. There’s no reward term for the signal: no explicit bonus for truthful signalling, no penalty for lying. This is a deliberate methodological commitment. If we rewarded truthful signalling directly, RQ1 would be answered by construction - separation would emerge because we paid for it. Keeping R_i free of any signal-dependent term ensures the regime we observe is an endogenous consequence of interaction structure, not reward engineering.

This means the signalling head receives gradient only indirectly, through the effect of its signal on neighbour behaviour, which in turn affects R_i^{drive} . It’s the same credit-assignment pattern that emergent-communication work [4] must solve - but with the added complexity that our messages also have direct physical consequences for the sender. We expect this credit-assignment difficulty to be the principal practical challenge of the approach.

3.5 Preference Divergence and Separating Equilibrium Threshold

The degree to which the two types face misaligned incentives is captured by a scalar preference divergence:

$$d = 1 - \frac{w_c \cdot w_a}{\|w_c\| \|w_a\|}, \quad (2)$$

where $w_\theta = (w_\theta^{\text{safety}}, w_\theta^{\text{progress}}, w_\theta^{\text{comfort}})$ is the reward-weight vector for type θ . $d = 0$ when both types share identical weights (the same-reward control); d increases as the weight vectors diverge.

The Crawford–Sobel / Spence single-crossing condition for a separating equilibrium requires that the reward differential of using profile m_1 versus m_3 has opposite signs across types. Define the profile differential vector $\Delta = (\Delta_s, \Delta_p, \Delta_k)$, where $\Delta_i = \mathbb{E}[r_i \mid m_1] - \mathbb{E}[r_i \mid m_3]$ is the expected per-step reward component under profile m_1 minus that under m_3 . From Table 1: $\Delta_s > 0$ (m_1 is safer), $\Delta_p < 0$ (m_1 is slower), and $\Delta_k > 0$ (m_1 is smoother). Separation is supportable iff

$$w_c \cdot \Delta > 0 \quad \text{and} \quad w_a \cdot \Delta < 0. \quad (3)$$

To derive the minimum divergence d_{sep} at which (3) first holds simultaneously, parameterise the weight vectors as

$$w_c(\alpha) = \bar{w} + \alpha \delta, \quad w_a(\alpha) = \bar{w} - \alpha \delta, \quad (4)$$

where $\bar{w} = \frac{1}{2}(w_c + w_a)$ is the mean weight vector and $\delta = \frac{1}{2}(w_c - w_a)$ is the half-difference; $\alpha = 0$ gives the same-reward baseline and $\alpha = 1$ recovers the current bimodal configuration. Each inequality in (3) imposes a linear threshold on α :

$$\alpha_c = \frac{-\bar{w} \cdot \Delta}{\delta \cdot \Delta}, \quad \alpha_a = \frac{\bar{w} \cdot \Delta}{\delta \cdot \Delta}, \quad \alpha_{\text{sep}} = \max(\alpha_c, \alpha_a). \quad (5)$$

Converting back to cosine distance gives $d_{\text{sep}} = 1 - \cos(w_c(\alpha_{\text{sep}}), w_a(\alpha_{\text{sep}}))$. If the current bimodal condition satisfies $d > d_{\text{sep}}$, the formulation lies in the analytically supportable separation regime. The d -sweep in Section 4.3 tests whether empirical ρ tracks this boundary across five divergence levels. Note that d and d_{sep} are our operationalisations of the preference-bias concept introduced by Crawford and Sobel [3]; the single-crossing condition in (3) is the discrete analogue of Spence’s [14] costly-signalling separability criterion applied to our reward structure.

3.6 Environment

The environment is **highway-env** [7], extended to support (i) heterogeneous agent populations drawn from a configurable type distribution, and (ii) the discrete execution-profile

vocabulary implemented as low-level controllers layered on the existing continuous action space. A single scenario is used throughout: dense multi-lane highway driving with $N = 6$ agents at nominal traffic density.

4 Evaluation Protocol

4.1 Primary Diagnostic

The central measurement is the empirical mutual information between the latent type and the emitted signal, estimated from evaluation rollouts:

$$I(\Theta; \mathcal{M}) = \sum_{\theta \in \Theta, m \in \mathcal{M}} p(\theta, m) \log \frac{p(\theta, m)}{p(\theta) p(m)}. \quad (6)$$

where $\Theta = \{\text{cautious}, \text{assertive}\}$ is the type space; $\mathcal{M} = \{m_1, m_2, m_3\}$ is the signal vocabulary; $p(\theta, m)$ is the empirical joint frequency of type θ emitting signal m , estimated from evaluation rollouts; $p(\theta)$ is the marginal type frequency (fixed at 0.5 each in the bimodal condition); $p(m)$ is the marginal signal frequency, aggregated across all types; and the log-ratio $\log \frac{p(\theta, m)}{p(\theta) p(m)}$ measures how much more often the pair (θ, m) co-occurs than expected under statistical independence - zero when type and signal are unrelated, positive when a type preferentially selects a signal.

We report $I(\Theta; \mathcal{M})$ in bits and normalise by $\log |\Theta|$ to give a separation coefficient $\rho \in [0, 1]$. Operationally: $\rho < 0.1$ indicates pooling (the signal carries essentially no information about type); $\rho > 0.9$ indicates separating (the signal determines type); intermediate values indicate partial-pooling. The distribution of ρ across seeds and conditions directly addresses RQ1 and RQ2.

Per-signal conditional distributions $p(\theta | m)$ are reported as a supplementary diagnostic, revealing whether any partial-pooling structure is clean - some signals fully informative, others uninformative - or gradual, where all signals carry intermediate information.

4.2 Intervention Test

Mutual information shows that signals are correlated with type but does not establish that receivers exploit those signals. A correlation - cautious agents happen to emit m_1 for kinematic reasons unrelated to any communicative act - could produce elevated ρ with no actual signalling function. To test causal use, the following seven conditions are evaluated on the best bimodal checkpoint, each over 200 episodes:

1. **Intact (baseline).** Unmodified policy; all ablations are compared against this reference on collision rate ($\Delta_{\text{collision}}$) and merge-success rate (Δ_{merge}).
2. **Signal randomisation.** Each emitted token is replaced with a uniform draw from $\mathcal{M}$ before being passed to the environment and to neighbour observations. If

receivers condition on token content, policy quality should degrade. ρ collapses by construction here; the relevant measure is driving-performance degradation.

3. **Signal hiding.** The signal component is zeroed out in each neighbour’s observation vector, so receivers see kinematics only with no token. This tests whether the response head uses the signal channel at all.
4. **Signal permutation.** The mapping from token to execution profile is scrambled, so the kinematic consequence of each token no longer matches what was learned. This separates the communicative token from its physical referent.
5. **Kinematic hiding.** Kinematic observations are zeroed while signal tokens are left intact. Receivers see tokens only, with no co-occurring kinematics. This tests whether tokens carry information independently of the kinematic context in which they are emitted.
6. **Full blackout.** Both signal tokens and kinematic observations are zeroed. This establishes floor performance when receivers are entirely blind, providing a lower bound against which the partial hiding conditions are interpreted.
7. **Desync.** The policy token is emitted and remains visible in neighbour observations as normal, but a uniformly random execution profile is applied kinematically at each step via a low-level override. This breaks the token $\leftrightarrow$ kinematic-image binding without altering what receivers observe in the token channel. A crash spike under desync - but not under signal hiding - indicates that receivers relied on the kinematic image of the token rather than the token itself; the absence of such a spike indicates that the token channel is the primary carrier. To our knowledge, no prior emergent-communication study includes this ablation: cheap-talk architectures have no separable physical-execution component, so the token-kinematic decoupling that desync exploits cannot arise.

The critical diagnostic is the comparison between conditions 3 and 7: elevated ρ that survives signal hiding ($\rho_{\text{hide}} \approx \rho_{\text{intact}}$) indicates the type-signal correlation is kinematic rather than communicative; a desync crash spike confirms that receivers had been exploiting the kinematic image of the token. Conditions 3–6 jointly decompose the signal channel into its token and kinematic components and establish whether each independently supports coordination.

4.3 Experimental Conditions

Three conditions are compared, each across twenty training seeds:

1. **Full method, bimodal population.** 50% cautious, 50% assertive. The principal condition; addresses RQ1.
2. **Full method, same-reward bimodal population.** Both type labels {cautious, assertive} are present and sampled equally (50% each), but both receive the same

reward weight vector - identical safety, progress, and comfort coefficients. The signalling head observes a type token, but the token carries no behavioural content. Tests RQ2: ρ should remain near the random-signal level here but be elevated in condition 1 if type-behavioural content, not the label itself, drives separation.

3. **Random-signal baseline, bimodal population.** Identical architecture with the signalling head replaced by uniform random emission over $\mathcal{M}$. Controls for the possibility that any measured ρ is an artefact of the extra action dimension rather than learned informativeness.

4.4 Statistical Protocol

Results are reported as per-seed values with means and 95% confidence intervals across twenty seeds. Evaluation uses 200 episodes per seed. Driving-performance metrics - collision rate, near-miss count, merge-success rate - are logged throughout for contextual reporting.

5 Expected Contributions

1. **Formulation.** A cooperative repeated signalling-game formulation of multi-agent highway driving with a behaviourally-grounded, non-cheap-talk signal space, closing the gap between analytical signalling-game lane-change models [13] and learning-based MARL.
2. **Operational diagnostic.** The separation coefficient ρ , grounded in empirical mutual information, as a method-independent criterion for classifying trained MARL populations as pooling, partial-pooling, or separating; complemented by a seven-battery intervention test — including a novel desync condition that keeps the emitted token visible to receivers while randomising the sender’s physical execution, an ablation that cannot be constructed in cheap-talk architectures — that moves beyond correlation to causal use of the signal channel.
3. **Experimental demonstration.** A three-condition experiment establishing that the formulation trains end-to-end, that ρ responds to the behavioural content of type labels rather than their mere presence, that the predominant pooling regime at the current preference divergence ($d = 0.30$, mean $\rho = 0.087$) is consistent with the analytically derived threshold d_{sep} — separation is not expected below this value — and that ρ is elevated 43-fold above the same-reward control across all seeds, attributing measured separation to the behavioural content of type labels rather than to architectural or labelling effects. A transformer encoder is identified as a future extension if the GRU baseline trains reliably.

6 Practical Relevance

Informal behavioural signalling is part of how human drivers coordinate - and the absence of a modelled signalling layer is one plausible source of the stilted or freezing interactions sometimes seen in deployed autonomous systems. A driving stack that can emit legible behavioural cues and read them is likely to interact more fluidly with human traffic, regardless of whether those specific signals emerge from training or are deliberately designed. This thesis provides the measurement apparatus needed to ask whether a given MARL driving policy has learned a legible behavioural vocabulary at all - which is a prerequisite for deliberate design.

References

- [1] Haoyu Bai, Shaojun Cai, Nan Ye, David Hsu, and Wee Sun Lee. Intention-aware online POMDP planning for autonomous driving in a crowd. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 454–460, 2015.
- [2] Long Chen, Yunjie Li, Chao Huang, Bai Li, Yang Xing, Daxin Tian, Li Li, Zhongxu Hu, Xiaoxiang Na, Chen Lv, et al. Milestones in autonomous driving and intelligent vehicles: Survey of surveys. *IEEE Transactions on Intelligent Vehicles*, 8(2):1046–1056, 2022.
- [3] Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50(6):1431–1451, 1982.
- [4] Jakob N. Foerster, Yannis M. Assael, Nando de Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 2137–2145, 2016.
- [5] Charles W. Fox, Fanta Camara, Gustav Markkula, Richard Romano, Ruth Madigan, and Natasha Merat. When should the chicken cross the road? - game theory for autonomous vehicle - human interactions. In *International Conference on Vehicle Technology and Intelligent Transport Systems*, 2018. URL <https://api.semanticscholar.org/CorpusID:14001964>.
- [6] Drew Fudenberg and Jean Tirole. *Game Theory*. MIT Press, Cambridge, MA, 1991.
- [7] Edouard Leurent. An environment for autonomous driving decision-making. <https://github.com/Farama-Foundation/HighwayEnv>, 2018.
- [8] Shuijing Liu, Peixin Chang, Haonan Chen, Neeloy Chakraborty, and Katherine Driggs-Campbell. Learning to navigate intersections with unsupervised driver trait inference. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 3576–3582, 2022.

- [9] Yuchen Liu and Shuzhen Diao. An automatic driving trajectory planning approach in complex traffic scenarios based on integrated driver style inference and deep reinforcement learning. *PLOS ONE*, 19(1):e0297192, 2024.
- [10] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 6379–6390, 2017.
- [11] Xiaobai Ma, Jiachen Li, Mykel J. Kochenderfer, David Isele, and Kikuo Fujimura. Reinforcement learning for autonomous driving with latent state inference and spatial-temporal relationships. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 6064–6071, 2021.
- [12] Malte Risto, Colleen Emmenegger, Erik Vinkhuyzen, Melissa Cefkin, and Jim Hollan. Human-vehicle interfaces: The power of vehicle movement gestures in human road user coordination. In *Proceedings of the 9th International Driving Symposium on Human Factors in Driver Assessment, Training, and Vehicle Design*, 2017.
- [13] Yi Shao, Yanjun Zheng, Yong He, Meiting Gao, and Jian Sun. A discretionary lane-changing decision-making mechanism incorporating drivers’ heterogeneity: A signalling game-based approach. *Journal of Advanced Transportation*, 2020:Article ID 8892693, 2020.
- [14] Michael Spence. Job market signaling. *The Quarterly Journal of Economics*, 87(3): 355–374, 1973.
- [15] Ziran Wang, Xishun Liao, Chao Wang, David Oswald, Guoyuan Wu, Kanok Bori-boonsomsin, Matthew J. Barth, Kyungtae Han, BaekGyu Kim, and Prashant Tiwari. Driver behavior modeling using game engine and real vehicle: A learning-based approach. *IEEE Transactions on Intelligent Vehicles*, 5(4):738–749, 2020.
- [16] Chao Yu, Akash Velu, Eugene Vinitzky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [17] Ekim Yurtsever, Jacob Lambert, Alexander Carballo, and Kazuya Takeda. A survey of autonomous driving: Common practices and emerging technologies. *IEEE Access*, 8:58443–58469, 2020.
- [18] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 3391–3401, 2017.
- [19] Bingyu Zhou, Wilko Schwarting, Daniela Rus, and Javier Alonso-Mora. Joint multi-policy behavior estimation and receding-horizon trajectory planning for automated urban driving. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 2388–2394, 2018.